# Enrico Castelli

*M.Sc. Student in Automation and Control Engineering*

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*chettante*
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## Profile

Master's student in Automation and Control Engineering at Politecnico di Milano (current weighted average **29/30**), with a strong interest in mobile robotics, perception and control. Hands-on experience in autonomous field robotics (AIRLab FRE team), computer vision, graph SLAM and embedded neural networks.

## Education

2025–present **M.Sc. in Automation and Control Engineering**, *Politecnico di Milano*, Milan, Italy, *Laurea Magistrale*, class LM-25
Enrolled 20/08/2025, 2nd year. Current weighted average: **29/30** (106.33/110), 60 ECTS passed. Planned thesis: 20 ECTS.

2022–2025 **B.Sc. in Automation Engineering**, *Politecnico di Milano*, Milan, Italy, Final grade: **100/110**

2017–2022 **Scientific High School Diploma**, *Liceo Scientifico "M. Allegretti"*, Vignola (MO), Italy, Final grade: **100/100**

## M.Sc. Transcript of Records

| Course | ECTS | Grade | Date |
|---|---|---|---|
| Dynamics of Mechanical Systems | 10 | 30/30 | 01/2026 |
| Model Identification and Data Analysis | 10 | 27/30 | 01/2026 |
| Computer Aided Manufacturing | 10 | 30/30 | 02/2026 |
| Advanced and Multivariable Control | 10 | 27/30 | 06/2026 |
| Automation and Control in Electric and Hybrid Vehicles | 5 | 30 cum laude | 06/2026 |
| Perception, Localization and Mapping for Mobile Robots | 5 | 30 cum laude | 06/2026 |
| Dynamics of Electrical Machines and Drives | 10 | 30/30 | 07/2026 |
| **Weighted average: 29/30 (106.33/110)** – 60 ECTS, as of 10/10/2026 | | | |

In progress Artificial Neural Networks and Deep Learning; Automation and Control in Autonomous Vehicles; Automation and Control Laboratory; Collaborative, Intelligent and Humanoid Robotics; Image Analysis and Computer Vision; Robotics Laboratory; Software Engineering (for Automation); Advanced Topics in Automation and Control Engineering; Thesis (20 ECTS).

## Research & Project Experience

03/2026–06/2026 **Member, AIRLab FRE Team (Field Robot Event)**, *AIRLab, Politecnico di Milano*, Milan, Italy
International autonomous agricultural robotics competition; took part in the 23rd Field Robot Event in Bernburg, Germany (16–18 June 2026). Developed marker recognition and localization using OpenCV histogram back-projection and graph SLAM within a ROS 2 system (project page and technical report).

10/2025–present **Member, AI-ARMS Team**, *Automation Engineering Association, Politecnico di Milano*, Milan, Italy

Built and teleoperated (leader–follower) an open-source, 3D-printed SO-101 robotic arm. Trained an ACT imitation-learning policy with LeRobot on 50 teleoperated demonstrations for autonomous pick-and-place without teleoperation (project page).

02/2025–06/2025 **Student Competition: range maximization of electric vehicles**, *Politecnico di Milano*, Milan, Italy

Extracurricular course: designed a control strategy to improve the energy efficiency of an RC car using MATLAB and Simulink; teamwork, communication and leadership in a project team.

## Additional Training

03/2026 **Elements of Neural Networks for Microcontrollers and Edge AI**, *Politecnico di Milano, taught by STMicroelectronics*

Trained image-recognition networks in Keras (Google Colab) and deployed them on an STM32 microcontroller with ST software tools.

## Certifications

07/2026 **Learn Object-Oriented Programming in C++**, *Educative*

Online course, 6 hours. Certification ID: G9UXRNM104

07/2026 **Learn C++: The Complete Course for Beginners**, *Educative*

Online course, 10 hours. Certification ID: 0812FINJCK

06/2026 **Certificate of Participation – 23rd Field Robot Event**, *Hochschule Osnabrück & TU Braunschweig*, Bernburg (Saxony-Anhalt), Germany

International field robotics competition, 16–18 June 2026, as a member of the Politecnico di Milano (AIRLab) team.

07/2025 **TOEIC Listening & Reading**, *ETS Global*

Score **980/990** (Listening 495, Reading 485), CEFR level C1. Valid until 07/2027.

06/2025 **Open Badge – Student competition: control for minimizing the energy consumption of an electric vehicle**, *Politecnico di Milano*

Digital badge issued by Politecnico di Milano: view badge.

## Skills

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|---|---|
| Programming | MATLAB, Simulink, Python, C++ (incl. object-oriented programming), LaTeX |
| Robotics & Vision | ROS 2, OpenCV, graph SLAM, localization and mapping |
| Machine Learning | Imitation learning (LeRobot, ACT), Keras / TensorFlow, neural networks on microcontrollers (STM32) |
| Control | Multivariable control, system identification, electric drives control |

## Languages

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|---|---|---|
| Italian | Mother tongue | |
| English | C1 | *TOEIC Listening & Reading: 980/990* |

## Interests

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|---|---|
| Sport | Competitive basketball player since age 8 — teamwork, communication and a positive team environment |
| Other | Robotics, travel, trekking, skiing |

I authorize the processing of my personal data pursuant to Regulation (EU) 2016/679 (GDPR) and Italian Legislative Decree 196/2003.